

Pulkit Kumar

+91 9003037804 | pulkit.talks@gmail.com | buddywhitman.tech | [LinkedIn](#) | [GitHub](#) | [ORCID](#)

Pre-final year undergrad with experience in distributed enterprise software, production RAG, recommendation engines & agentic AI, and edge-optimized SSR applications. Deployed high-throughput infrastructure serving a cumulative 250K+ users across 8+ domains to date.

TECHNICAL SKILLS

Languages: Python, Java, TypeScript, Go, C/C++, JavaScript, SQL, HTML5, CSS
AI/ML: PyTorch, TensorFlow, Keras, LangGraph, LangChain, OpenCV, Scikit-learn, LiveKit, NLTK
Backend: REST, FastAPI, Spring Boot, Node.js, GraphQL, Django, Flask, Celery, Kafka, Redis, BullMQ
Frontend: Next.js, React, Angular, Vue, Serverless (CloudFlare/AWS), Tailwind, Flutter, Bun, Vite, Babel
Cloud: AWS (ECS, Fargate, Bedrock, Aurora, RDS, Lambda), Terraform, GCP (Cloud Run, BigQuery, Vertex)
Databases: PostgreSQL (pgvector), MySQL, Milvus, Pinecone, MongoDB, DynamoDB, Spanner, InfluxDB
DevOps: Docker, Kubernetes, Git, GitHub Actions, Jenkins, Prometheus, Grafana, ELK, OpenTelemetry

EXPERIENCE

Software Development Engineer January 2025 – Present
Internship - Fettle *Hyderabad, IN*

- Implemented a cross-modal hybrid recommender engine using PyTorch & TorchRec, combining real-time wearable/bio-signal embeddings with collaborative filtering for 20K+ users
- Built an agentic AI orchestration layer using Google ADK to bridge natural language interfaces with enterprise ERPs, reducing manual workflows with NLU-driven task execution across 17+ client accounts

Applied AI Engineer July 2025 – December 2025
Internship - Teachafy Labs *Bengaluru, IN*

- Deployed RAG workflows for compliance intelligence using Pinecone + OpenAI Assistant API with semantic Redis caching, achieving 95% citation accuracy and reducing audit timelines by almost half
- Designed a fault-tolerant, Kubernetes HPA-centric document intelligence pipeline with distributed locking, observability and parallelism for stable multi-TB processing, cutting manual employee costs by 80%

Dev Team Lead December 2024 – October 2025
Student Council - Manipal Institute of Technology *Manipal, IN*

- Developed distributed ticketing infrastructure (50K MAU) on AWS ECS/Fargate with autoscaling, sustaining 10x burst loads while maintaining 99.99% uptime reliability during peak booking windows
- Optimized global delivery using AWS CDN & Edge Workers, reducing TTFB by 250ms and improving core web vitals (FCP/LCP by 70%), which was validated under JMeter load simulations (100k+ hourly requests)

EDUCATION

Manipal Institute of Technology Manipal, India
Bachelor of Technology in Electrical & Electronics Engineering *August 2023 – July 2027 (Expected)*

PROJECTS

Solar Racing Digital Twin Engine | *Kafka, Spark, Zookeeper, Hadoop, Grafana, CasADi, Ipopt* Jun 2024

- Architected Lambda Architecture with Spark Streaming and Kafka (synced via Zookeeper) to fuse high-frequency CAN telemetry (> 500Hz) with geospatial APIs for sub-100ms state synchronization
- Filed **2 patents** for this and an NMPC-driven energy optimizer using HDFS data lakes and physics-informed LSTMs to solve multi-objective constraints on TB-scale telemetry and historical race data

American Regulatory Intelligence | *FastAPI, Next, LegalBERT, Playwright, Redis, Celery, AWS* Dec 2025

- Built event-driven microservices to ingest and monitor regulatory updates across Federal Register, eCFR and 50+ US state agencies, achieving <15-minute time-to-insight
- Designed an inference layer for 95%+ topic classification accuracy and RAG summarization with HITL controls, implementing heuristic impact scoring and semantic diff for policy evolution tracking